

Inductive Bias in Retrieval

A Multi-Model, Multi-Dataset Study

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Project Idea

**Does the inductive bias of a retriever determine
its performance on biomedical retrieval?**

Or does the right answer depend on the dataset and query type?

Bias class	Example model
Lexical sparse	BM25
General-domain dense	MiniLM, BGE-small, E5-small
Learned sparse	SPLADE
Domain-adapted dense	MedCPT (PubMed-trained)

6 retrievers · 5 BEIR-style benchmarks · 3 add-ons (cross-encoder reranker, per-query router, downstream RAG)

Theoretical Framing

Three course concepts anchor the analysis.

1 No Free Lunch

No inductive bias is universally optimal. Performance depends on alignment between the bias and the data distribution.

2 Empirical vs. true risk

How well does the test-set estimate of the loss approximate the expected risk on the full query distribution?

3 Stability

Does the system ranking hold under different query subsets, and across different datasets?

Inferential tools: paired bootstrap CI · Hoeffding sample-size bound · train–test generalisation gap

What Was Built

Dataset	BM25	MiniLM	BGE	E5	SPLADE	MedCPT
NFCorpus	✓	✓	✓	✓	✓	✓
BioASQ-subset	✓	✓	✓	✓	✓	✓
TREC-COVID	✓	✓	✓	✓	—	—
SciFact	✓	✓	✓	✓	✓	✓
ArguAna	✓	✓	✓	✓	✓	✓

Three additions on top of the first stage:

- BGE cross-encoder reranker, plus in-domain MedCPT cross-encoder
- Per-query router (logistic + LightGBM, six query features)
- Downstream PubMedQA QA with flan-t5-base (with retrieval ablation)

All experiments fit in a single Google Colab T4 session.

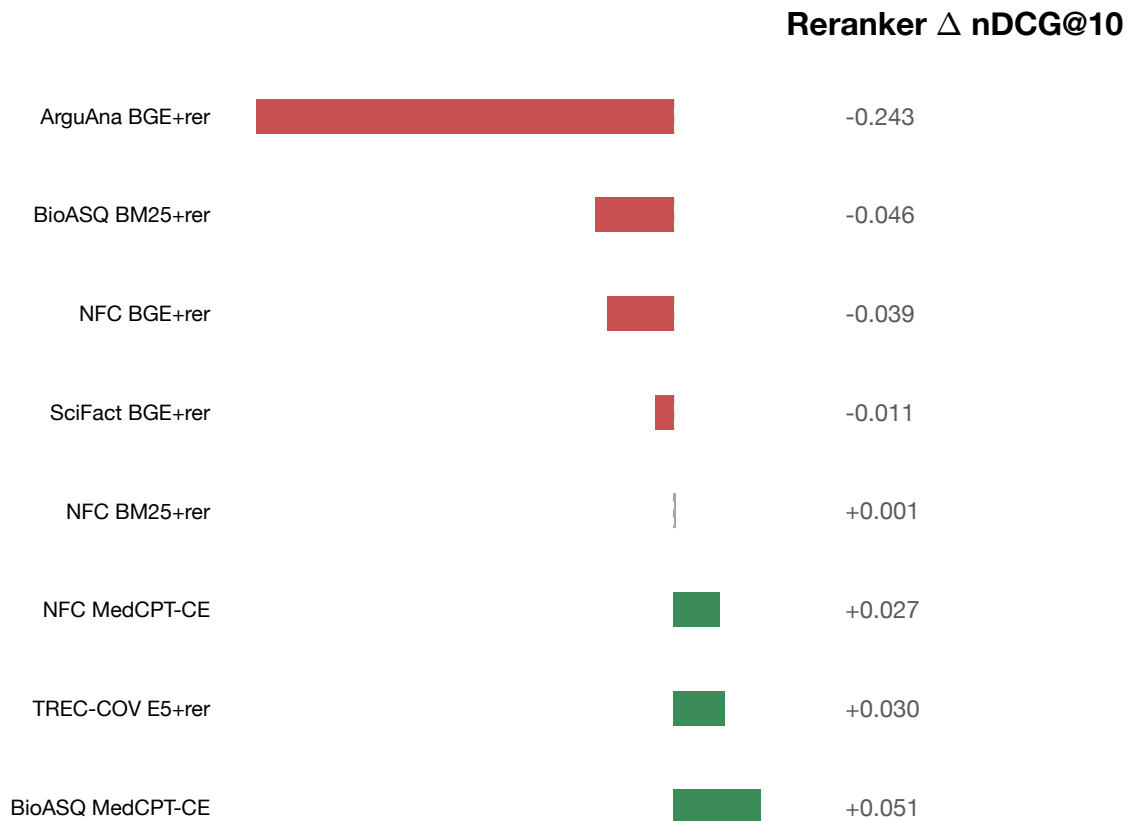
Result 1: No Single Winner

nDCG@10

Dataset	BM25	MiniLM	BGE	E5	SPLADE	MedCPT
NFCorpus	0.294	0.317	0.339	0.327	0.333	0.325
BioASQ-subset	0.754	0.539	0.684	0.666	0.673	0.610
TREC-COVID	0.576	0.454	0.645	0.724	—	—
SciFact	0.662	0.645	0.720	0.688	0.633	0.710
ArguAna	0.361	0.370	0.429	0.310	0.394	0.133

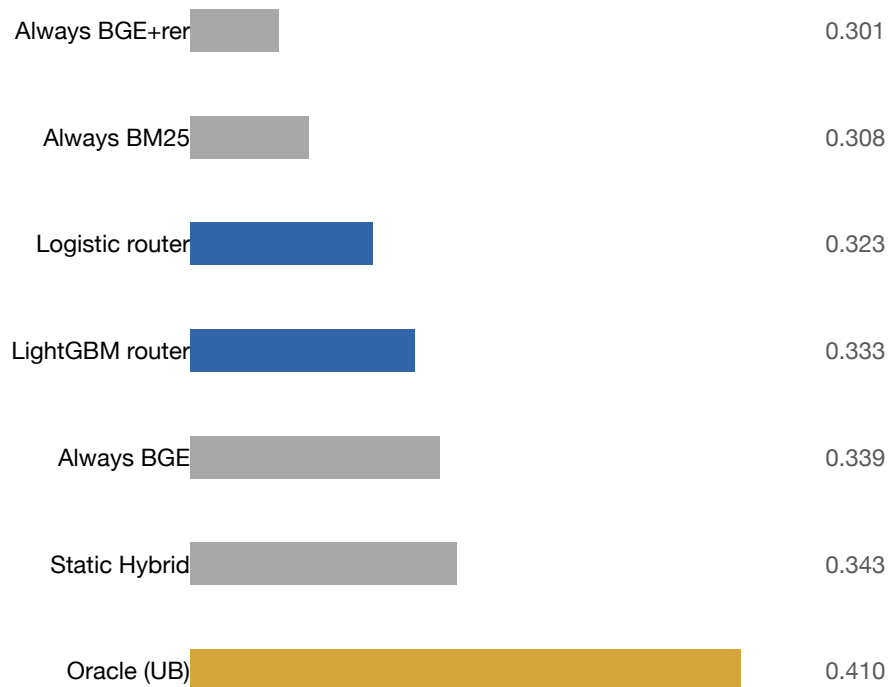
- BGE-small wins 3 datasets, E5-small wins 1, BM25 wins 1.
- MedCPT collapses on ArguAna (−0.227 vs. BM25): biomedical pre-training actively hurts on a non-biomedical task.
- BioASQ-subset corpus has zero distractors → BM25 inflated; treat as control, not as biomedical evidence.

Result 2: Stronger Reranker $\neq$ Stronger Pipeline



*BGE-rerank hurts on 4 of 5 datasets.
Domain-aligned MedCPT-CE wins on biomedical sets.*

Router results (NFCorpus)



$\Delta(q) = \text{nDCG@10}_{\text{Dense}}(q) - \text{nDCG@10}_{\text{BM25}}(q)$, the per-query advantage of Dense over BM25 on NFCorpus. An OLS regression on three standardised query features finds two significant predictors — **vocabulary gap** ($\beta = +0.048$, $p < 0.001$) raises Δ ; **technicality / mean IDF** ($\beta = -0.034$, $p = 0.019$) lowers it; query length is not significant. $R^2 = 0.071$. The router uses these three features plus three more (BM25 top-1 score, question-form flag, medical-token share) to predict the best strategy per query.

Take-Aways and Limitations

Three findings:

- 1. Inductive bias is dataset-specific.** No retriever and no reranker dominate the five datasets.
- 2. Domain alignment matters at the cross-encoder stage**, not at the bi-encoder stage. MedCPT-CE gives the only clean biomedical gains in the study.
- 3. Per-query routing is now a method.** The signal is real (oracle +0.066 above the best static fusion) but six features capture only $\sim 7\%$ of $\Delta(q)$ variance.

Limitations: BioASQ-subset corpus has zero distractors (degenerate);